dimensions. In this manner, string theory, which appeared to be on the brink of elimination from the realm of physical relevance, is saved. Moreover, rather than just postulating the existence of extra dimensions, as had been done by Kaluza, Klein, and their followers, string theory requires them. For string theory to make sense, the universe should have nine space dimensions and one time dimension, for a total of ten dimensions. In this way, Kaluza's 1919 proposal finds its most convincing and powerful forum.

Some Questions

This raises a number of questions. First, why does string theory require the particular number of nine space dimensions to avoid nonsensical probability values? This is probably the hardest question in string theory to answer without appealing to mathematical formalism. A straightforward string theory calculation reveals this answer, but no one has an intuitive, nontechnical explanation for the particular number that emerges. The physicist Ernest Rutherford once said, in essence, that if you can't explain a result in simple, nontechnical terms, then you don't really understand it. He wasn't saying that this means your result is wrong; rather, he was saying that it means you do not fully understand its origin, meaning, or implications. Perhaps this is true regarding the extradimensional character of string theory. (In fact, let's take this opportunity to brace —parenthetically—for a central aspect of the second superstring revolution that we will discuss in Chapter 12. The calculation underlying the conclusion that there are ten spacetime dimensions—nine space and one time—turns out to be approximate. In the mid-1990s, Witten, based on his own insights and previous work by Michael Duff from Texas A&M University and Chris Hull and Paul Townsend from Cambridge University, gave convincing evidence that the approximate calculation actually misses one space dimension: String theory, he argued to most string theorists' amazement, actually requires ten space dimensions and one time dimension, for a total of eleven dimensions. We will ignore this important result until Chapter 12, as it will have little direct bearing on the material we develop before then.)

Second, if the equations of string theory (or, more precisely, the approximate equations guiding our pre-Chapter 12 discussion) show